



Chapter 9- Windows Registry

◆ What is Windows Registry?

The **Windows Registry** is a **central database** that stores **configuration settings** and **options** for:

- Microsoft Windows OS 
- Built-in Windows applications 
- Third-party applications 

👉 In simple words:

Registry = **“Brain of Windows Settings”**



Real-Life Example:

When you set your default browser to Chrome, that setting is stored in the registry. Next time you open a link, Windows already knows which app to use.

◆ How to Open Registry Editor (regedit)

You can open Registry Editor in 2 ways:

1. Start Menu → Search **“Registry Editor”**
2. Press **Windows + R** → Type **regedit** → Press Enter



Important Warning

Editing registry incorrectly can cause system issues or crashes.



Best Practice:

- Always take a backup before making changes
 - Use tools like **VMware** for testing 
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◆ Main 5 Registry Hives (Folders)

1 HKEY_CLASSES_ROOT (HKCR)

📍 Path: Computer\HKEY_CLASSES_ROOT

📌 **Purpose:**

Stores **file associations and default applications**

💡 **What it does:**

- Defines which application opens which file type
- Contains all file extensions (.docx, .pdf, .mp4, etc.)

💡 **Example:**

- .docx → opens in Microsoft Word
- .mp4 → opens in VLC Media Player

👉 If you double-click a file and it opens in the correct app, that mapping comes from HKCR.

2 HKEY_CURRENT_USER (HKCU)

📍 Path: Computer\HKEY_CURRENT_USER

📌 **Purpose:**

Stores settings for the **currently logged-in user**

💡 **What it includes:**

- Desktop background 🎨
- Theme settings 🌙
- Personal preferences

💡 **Example:**

If you change wallpaper, only your account sees it—not other users.

3 HKEY_LOCAL_MACHINE (HKLM)

📍 Path: Computer\HKEY_LOCAL_MACHINE

📌 Purpose:

Stores **system-wide settings** (applies to all users)

📁 Important Subfolders:

- **SAM** → Stores user account data (username & password info) 🔒
- **SYSTEM** → Windows OS configuration ⚙️
- **SOFTWARE** → Installed software settings 📦

💡 Example:

If you install software like Chrome, its configuration is stored here.

👉 Any changes here affect **all users**

4 HKEY_USERS (HKU)

📍 Path: Computer\HKEY_USERS

📌 Purpose:

Stores settings for **all users on the system**

💡 Use Case:

If multiple users exist and you want to edit settings for another user, you use this.

5 HKEY_CURRENT_CONFIG (HKCC)

📌 Path: Computer\HKEY_CURRENT_CONFIG

📌 **Purpose:**

Stores **current hardware configuration**

💡 **Example:**

- Display settings
- Connected devices

👉 Represents current system state (temporary configuration)

◆ Practical: Remove “Virus & Threat Protection” Icon

📌 Path:

HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows Defender Security Center\Virus and threat protection

🔧 **Steps:**

1. Open Registry Editor
2. Navigate to the above path
3. Right-click → New → DWORD (32-bit) Value
4. Name it: **UILockdown**
5. Set value:
 - 1 → Enable (Hide UI)
 - 0 → Disable
6. Restart system 🔄

⚠️ **Important:**

Always test in **VMware** to avoid damaging your main system.

◆ Useful Registry Resource Websites

- <https://www.tenforums.com/>
- <https://www.elevenforum.com/>

👉 These sites provide:

- Ready registry tweaks
- Troubleshooting solutions
- Windows 10 & 11 configurations

◆ Run & RunOnce Registry Keys (Auto Startup Programs)

These keys allow programs to run automatically when a user logs in.

✦ Run vs RunOnce

Key	Behavior
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Run	Runs every time user logs in 
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RunOnce	Runs only once, then deletes itself 
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📌 Registry Paths:

For All Users (System-wide):

HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\Run

HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\RunOnce

For Current User:

HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Run

HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\RunOnce

💡 Important Points:

- Command length limit: **260 characters**
- Format:

Description = CommandLine

- Multiple programs can be added
 - Execution order is **not fixed**
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⚠️ Malware Use Case (Very Important for IT Support)

Attackers use these registry paths to **auto-start malware**

👉 Why?

They want malware to run **before antivirus starts**

🔍 How to Detect Suspicious Entries:

Check if path includes:

- AppData
- ProgramData

👉 These are commonly used by malware

🔧 Troubleshooting Steps:

1. Open Registry → Go to Run/RunOnce
 2. Check unknown entries
 3. Copy file path
 4. Upload file to **VirusTotal.com** 🧪
 5. Analyze if malicious
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Real-Life Scenario:

If a system slows down on startup, check Run keys → disable unwanted apps → improves performance ⚡

TASK 1: Grey Out Date & Time Settings

Goal:

Disable “Set time automatically” (make it greyed out)

Registry Path:

HKEY_LOCAL_MACHINE\SOFTWARE\Policies\Microsoft\Windows\DateTime

Steps:

1. Open Registry Editor

2. Navigate to:

HKLM\SOFTWARE\Policies\Microsoft\Windows

3. Right-click → New → Key → Name: **DateTime**

4. Inside DateTime:

- Right-click → New → DWORD (32-bit)
- Name: **DisableAutoTimeSet**

5. Set value:

- 1 → Disable (Grey out setting)
- 0 → Enable

6. Restart system 

Use Case:

- Company laptops where users should not change time
 - Prevent syncing issues in domain environments
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TASK 2: Disable / Restrict Start Menu

Goal:

Control or restrict Start Menu access

Registry Path:

HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Policies\Explorer

Steps:

1. Open Registry Editor
 2. Navigate to above path
 3. If **Explorer key not present**:
 - Right-click → New → Key → Name: Explorer
 4. Inside Explorer:
 - Right-click → New → DWORD (32-bit)
 - Name: **NoStartMenuMorePrograms**
 5. Set value:
 - 1 → Restrict Start Menu
 - 0 → Enable normally
 6. Restart system
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Alternative Restriction Options:

Value Name	Purpose
NoStartMenuMorePrograms	Restrict programs list
NoViewContextMenu	Disable right-click
NoTrayContextMenu	Disable taskbar menu

Real-Life Use Case:

- Office PCs where users should not install or access apps
 - School computers 
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IT Support Mindset (Very Important)

How to Troubleshoot Fast:

1. Identify problem 
 2. Check related registry path
 3. Look for abnormal entries
 4. Compare with working system
 5. Test fix in safe environment (VMware)
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Pro Tips:

- Always backup registry before editing
 - Use Run keys to detect malware quickly
 - Learn common registry paths = faster troubleshooting
 - Use trusted websites for registry tweaks
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✓ Summary

- Registry = Core configuration system
 - 5 main hives store different types of data
 - Run/RunOnce = auto-start programs (important for malware detection)
 - Registry tweaks can control system behavior
 - Always test in VM before applying on real system
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Chapter 9 – Windows Registry (SUMMARY)

What is Windows Registry?

 Registry = Central database of Windows

Stores:

- OS settings 
- Application settings 
- User preferences 

 Simple:

 **Registry = Brain of Windows**

How to Open Registry

- Start Menu → Registry Editor
 - Windows + R → regedit
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Important Warning

- ✗ Wrong change → System crash
- ✓ Always take backup
- ✓ Test in VM

5 MAIN REGISTRY HIVES (VERY IMPORTANT)

HKCR (HKEY_CLASSES_ROOT)

 File associations

 Example:

- .mp4 → VLC
- .docx → Word

HKCU (HKEY_CURRENT_USER)

 Current user settings

 Example:

- Wallpaper
- Theme

HKLM (HKEY_LOCAL_MACHINE)

 System-wide settings

 Important:

- SAM → User data 
- SOFTWARE → Apps
- SYSTEM → OS

HKU (HKEY_USERS)

 All users settings

5 HKCC (HKEY_CURRENT_CONFIG)

👉 Current hardware configuration

🔄 Run & RunOnce (VERY IMPORTANT 🔥)

📌 Paths:

👉 For All Users:

- HKLM → Run / RunOnce

👉 For Current User:

- HKCU → Run / RunOnce
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💡 Difference:

Key	Behavior
Run	Runs every login 🔄
RunOnce	Runs once ⌚

⚠️ Malware Concept

👉 Malware uses Run keys to auto-start

🔍 Check:

- AppData
 - ProgramData
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🧠 Practical Use Cases

🔧 Disable Time Settings

👉 DisableAutoTimeSet = 1

🔧 Restrict Start Menu

👉 NoStartMenuMorePrograms = 1

🎯 CONCLUSION

👉 Core understanding:

- Registry = System brain 🧠
- Run keys = Startup control 🔁
- HKLM = System
- HKCU = User

👉 Final line:

“Registry knowledge = Advanced troubleshooting power” 🔥

Q & A

◆ BASIC REGISTRY QUESTIONS

? Q1: What is Windows Registry?

👉 Answer:

Windows Registry is a central database that stores configuration settings for the operating system, hardware, and applications. It helps Windows understand how to behave and how to interact with hardware and software.

? Q2: Why registry is important?

👉 Answer:

Because it stores all system settings. Without registry, Windows cannot function properly.

? Q3: How do you open registry?

👉 Answer:

Press Windows + R → type regedit.

◆ HIVES QUESTIONS (VERY IMPORTANT)

? Q4: What are registry hives?

👉 Answer:

Registry hives are main sections (folders) that store different types of configuration data.

? Q5: What is HKLM?

👉 Answer:

Stores system-wide settings that apply to all users.

? Q6: What is HKCU?

👉 Answer:

Stores settings for the currently logged-in user.

? Q7: What is HKCR?

👉 Answer:

Stores file associations and default application mappings.

◆ RUN KEYS QUESTIONS (VERY IMPORTANT 🔥)

? Q8: What is Run key?

👉 Answer:

Run key contains programs that automatically start when user logs in.

? Q9: Difference between Run and RunOnce?

👉 Answer:

- Run → Executes every login
 - RunOnce → Executes only once
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? Q10: Why attackers use Run key?

👉 Answer:

To automatically start malware whenever system boots.

◆ MALWARE & SECURITY QUESTIONS

? Q11: How to detect malware using registry?

👉 Answer:

- Check Run/RunOnce keys
 - Look for unknown entries
 - Check paths like AppData
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? Q12: Why AppData path suspicious?

👉 Answer:

Because malware hides there and runs without admin permission.

◆ PRACTICAL QUESTIONS

? Q13: System slow on startup, what will you check?

👉 Answer:

- Check Run keys
 - Disable unnecessary startup entries
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? Q14: How to remove suspicious startup program?

👉 Answer:

- Open registry
 - Go to Run key
 - Delete suspicious entry
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? Q15: How to backup registry?

👉 Answer:

- Open regedit
 - File → Export
 - Save backup
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◆ REAL JOB SCENARIOS

? Q16: Unknown program starts automatically?

👉 Answer:

👉 Check:

- Run keys
 - Startup apps
 - 👉 Remove entry
-

? Q17: System behaving abnormally?

👉 Answer:

- Compare registry with working system
 - Check abnormal entries
-

? Q18: User settings not applying?

👉 Answer:

👉 Check HKCU

◆ ADVANCED QUESTIONS

? Q19: What is registry corruption?

👉 Answer:

When registry entries are damaged or missing, causing system issues.

? Q20: How to fix registry issues?

👉 Answer:

- Restore backup
 - Use system restore
 - Fix specific keys
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◆ INTERVIEW IMPRESSION QUESTIONS

? Q21: What is your registry troubleshooting approach?

👉 Answer:

1. Identify issue
 2. Find related registry path
 3. Check abnormal entries
 4. Backup
 5. Apply fix
 6. Test
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